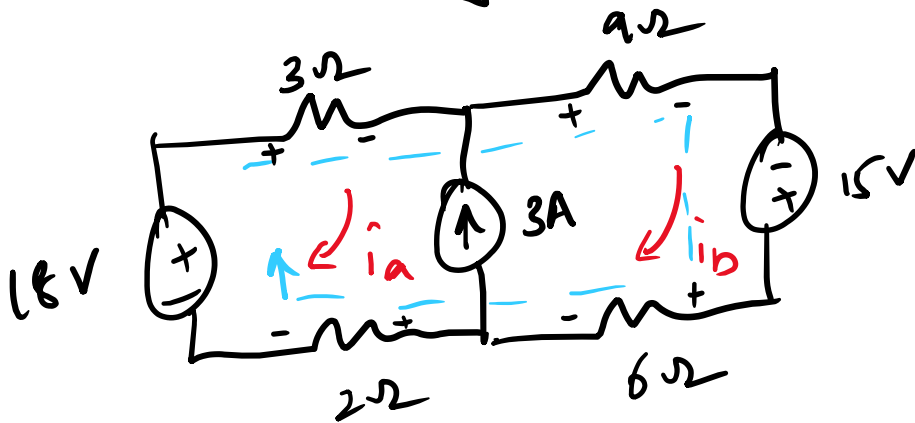


Mesh Analysis with a current source



Supermesh.

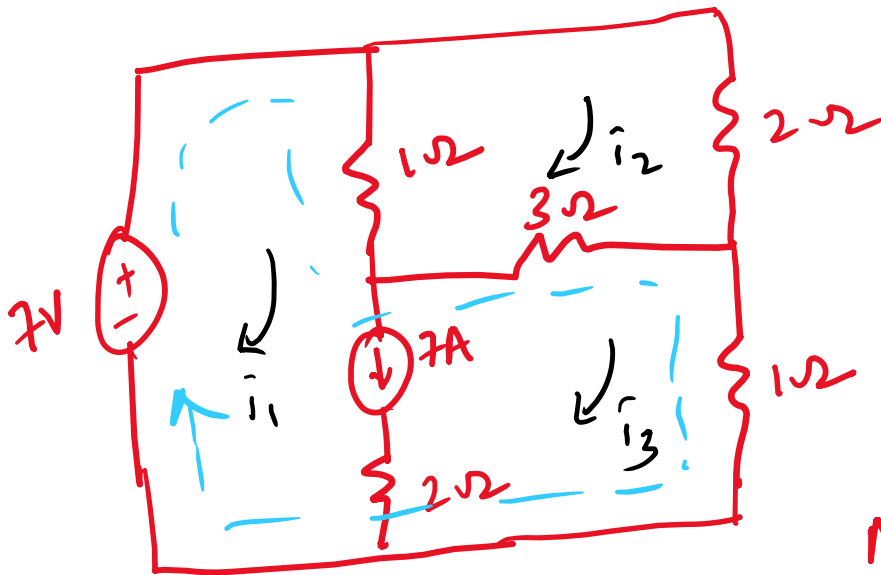
$$i_b - i_a = 3A$$

KVL Supermesh:

$$18 - 3i_a - 9i_b + 15 - 6i_b - 2i_a = 0$$

$$i_a = -0.6A$$

$$i_b = 2.4A$$



Find all the mesh currents.

$$\hat{i}_1 - \hat{i}_3 = 7A$$

Mesh 2 :

$$-1(\hat{i}_2 - \hat{i}_1) - 2\hat{i}_2 - 3(\hat{i}_2 - \hat{i}_3) = 0$$

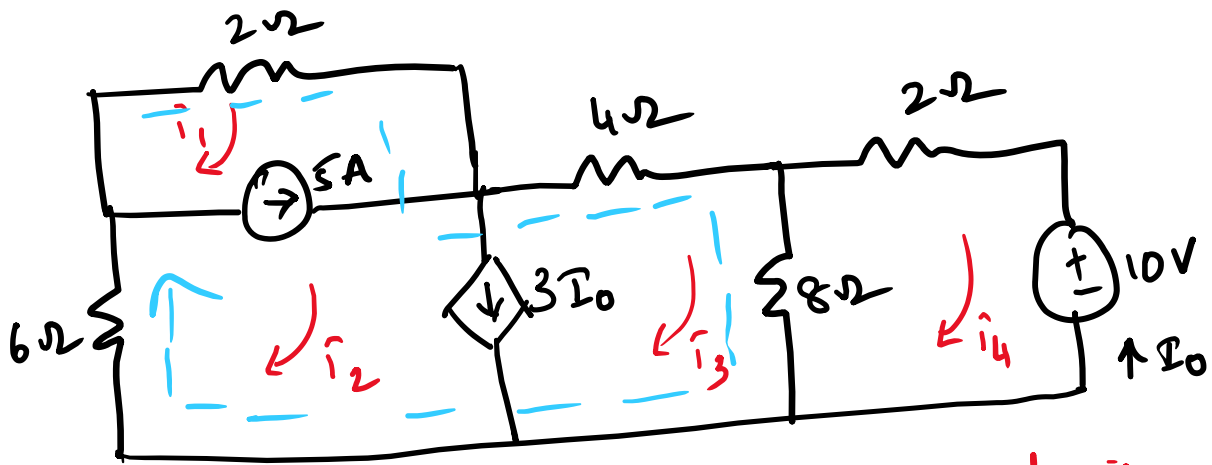
Supermesh :

$$7 - 1(\hat{i}_1 - \hat{i}_2) - 3(\hat{i}_3 - \hat{i}_2) - 1(\hat{i}_3) = 0$$

$$\hat{i}_1 = 9A$$

$$\hat{i}_2 = 2.5A$$

$$\hat{i}_3 = 2A$$



Find i_1 to i_4 using mesh analysis.

$$I_0 = -i_4$$

$$i_2 - i_1 = 5$$

$$i_2 - i_3 = -3i_4$$

Mesh 4:

$$-10 - 8(i_4 - i_3) - 2i_4 = 0$$

Supermesh: $-6i_2 - 2i_1 - 4i_3 - 8(i_3 - i_4) = 0$

$$i_1 = -7.5A$$

$$i_3 = 3.93A$$

$$i_2 = -2.5A$$

$$i_4 = 2.14A$$

Superposition

Voltage across (or current through) an element in a linear circuit is the algebraic sum of voltages across (or currents through) that element due to each independent source.

1. Turn off all independent sources except one. Find the output (V or I) due to the active source.
2. Repeat for all independent sources.
3. Find the total contribution by adding algebraically all the individual contributions.